

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

The focal length f is related to the radius of curvature r of the spherical convex mirror by -

(1) $f = r$

(2) $f = -\frac{1}{2}r$

(3) $f = +\frac{1}{2}r$

(4) $f = -r$

Q2: 25 Feb (Shift 1) - Numerical

The same size images are formed by a convex lens when the object is placed at 20 cm or at 10 cm from the lens. The focal length of convex lens is _____ cm.

Q3: 26 Feb (Shift 1) - Single Correct

A short straight object of height 100 cm lies before the central axis of a spherical mirror whose focal length has absolute value $|f| = 40$ cm. The image of object produced by the mirror is of height 25 cm and has the same orientation of the object. One may conclude from the information :

(1) Image is real, same side of concave mirror.

(2) Image is virtual, opposite side of convex mirror.

(3) Image is virtual, opposite side of concave mirror.

(4) Image is real, same side of convex mirror.

Q4: 26 Feb (Shift 2) - Single Correct

The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively. Then choose the correct relation for these vectors.

(1) $\vec{b} = 2\vec{a} + \vec{c}$

(2) $\vec{b} = \vec{a} - \vec{c}$

(3) $\vec{b} = \vec{a} + 2\vec{c}$

(4) $\vec{b} = \vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c}$

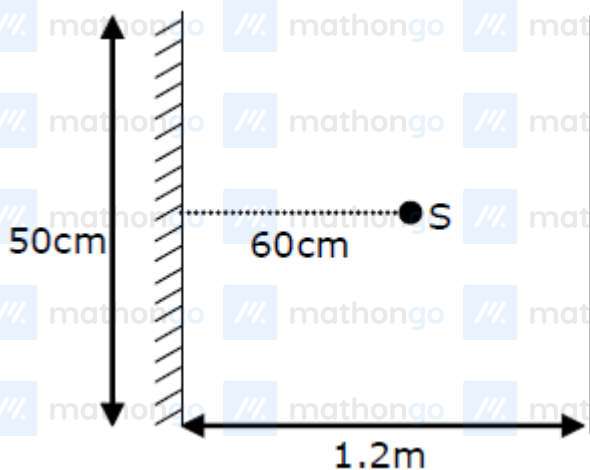
Q5: 26 Feb (Shift 2) - Single Correct

Given below are two statements: One is labeled as Assertion A and the other is labeled as Reason R . Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image. Reason R : Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both A and R are true but R is NOT the correct explanation of A
- (2) Both A and R are true and R is the correct explanation of A
- (3) A is true but R is false
- (4) A is false but R is true

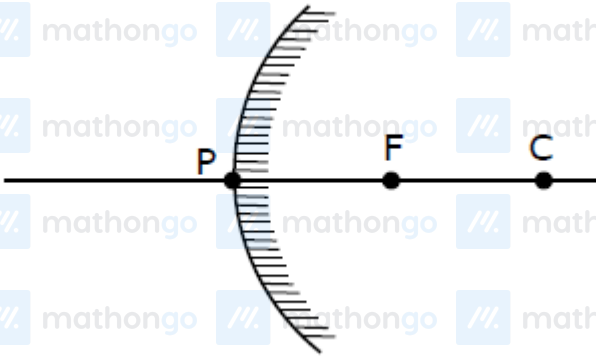
Q6: 26 Feb (Shift 2) - Numerical

A point source of light S , placed at a distance 60 cm in front of the centre of plane mirror of width 50 cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2 m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is cm



Answer Key**Q1 (3)****Q2 (15)****Q3 (2)****Q4 (4)****Q5 (2)****Q6 (150)**

Q1 (3)



$$\text{So, } \frac{R}{2} = f$$

$$F = +\frac{1}{2}R$$

Q2 (15)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \dots (1)$$

$$m = \frac{v}{u} \dots (2)$$

$$\text{from (1) and (2) we get } m = \frac{f}{f+u}$$

given conditions

$$m_1 = -m_2$$

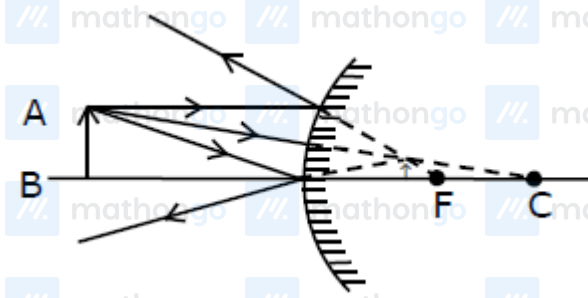
$$\frac{f}{f-10} = \frac{-f}{f-20}$$

$$f - 20 = -f + 10$$

$$2f = 30$$

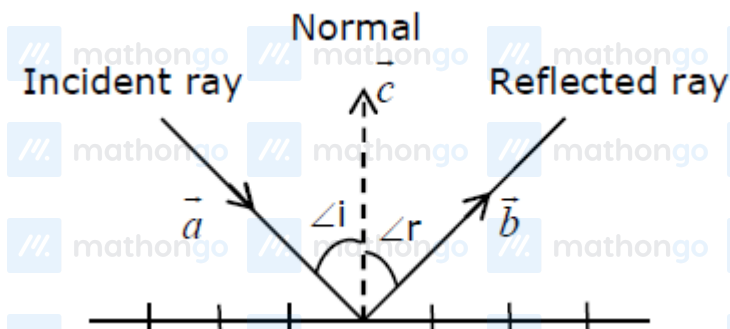
$$f = 15 \text{ cm}$$

Q3 (2)



Same orientation so image is virtual. It is combination of real object and virtual image using height it is possible only from convex mirror.

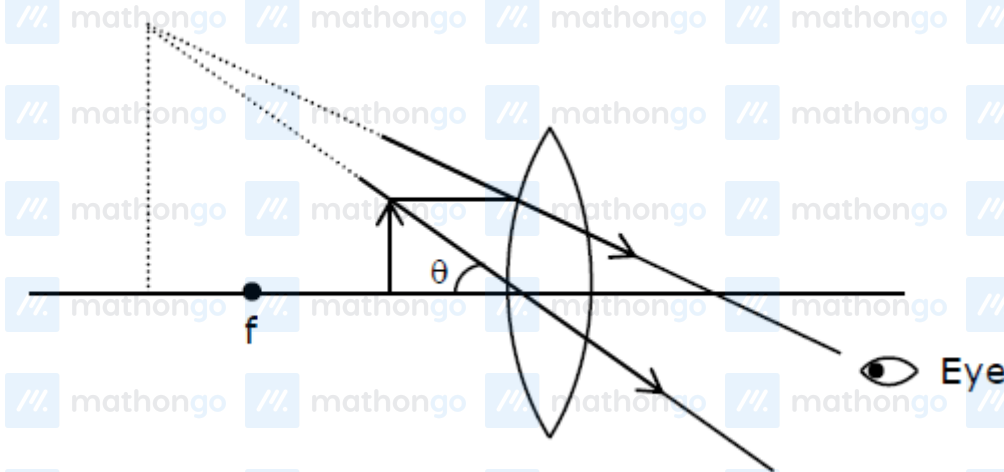
Q4 (4)



We see from the diagram that because of the law of reflection, the component of the unit vector $\vec{a}$ along $\vec{b}$ changes sign on reflection while the component parallel to the mirror remain unchanged. $\vec{a} = \vec{a}_{\parallel} + \vec{a}_{\perp}$ and $\vec{a}_{\perp} = \vec{c}(\vec{a} \cdot \vec{c})$

we see that the reflected unit vector is $\vec{b} = \vec{a}_{\parallel} - \vec{a}_{\perp} \Rightarrow \vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c}$

Q5 (2)

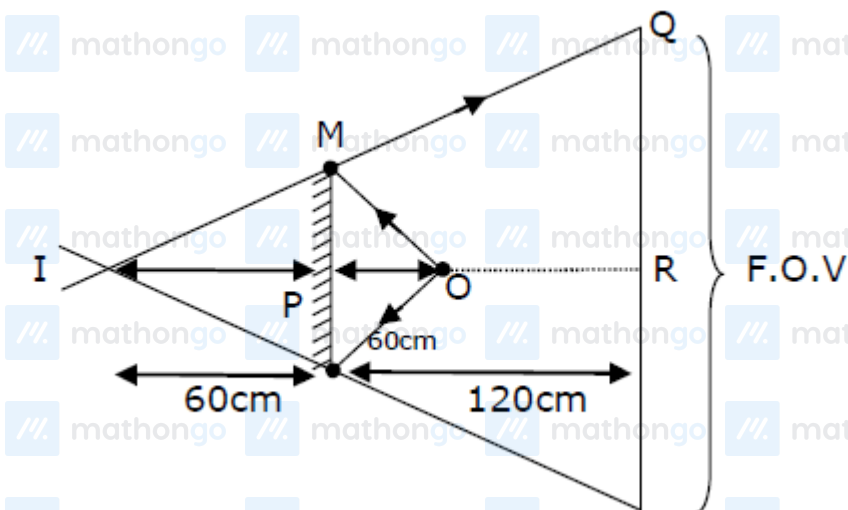


Both obtain same angle, since image can be at a distance greater than 25 cm, object can be moved closer to eye.

Q6 (150)

from similar triangle IMP and IQR

$$\frac{QR}{25} = \frac{180}{60} \Rightarrow QR = 7$$



$$\text{F.O.V.} = 2 \times 75 \Rightarrow 150^\circ$$